Mar. 17

- Working toward a proof of Lagrange's Theorem

  index of H in G

  - Thm: Let G be a finite group and H a subgroup. Write $[G:H]$ for the number of distinct (left or right) cosets of H and G. Then $|G| = |H| \cdot [G:H]$.
  - Corollary: $|H|$ divides $|G|$ for any subgroup $H \leq G$.

Ex. $G = S_3$, $H = A_3 = \langle (123) \rangle = \{id, (123), (132)\}$

The number of cosets of H in S_3 should be 2:

$|S_3| = 6$, $|H| = |\langle (123) \rangle| = 3$, $|S_3| = |A_3| \cdot [S_3 : A_3] \Rightarrow 6 = 3 \cdot [S_3 : A_3] \Rightarrow [S_3 : A_3] = 2$

Cosets: $e\langle (123) \rangle = \{e, (123), (132)\}$

$(12)\langle (123) \rangle = \{(23), (12), (13)\}$

exclusive or

- Lemma 1 Let $a, b \in G$. Then either $aH = bH$ or $aH \cap bH = \emptyset$ Two cosets are either equal or disjoint

  Proof Let aH, bH be cosets. If they are disjoint (i.e $aH \cap bH = \emptyset$), then we are done. Otherwise, suppose $\exists c \in aH \cap bH$. Thus $c = ah_1 = bh_2$ for some $h_1, h_2 \in H$. Let $ah_3 \in aH$ be arbitrary. We see $a = bh_2h_1^{-1}$. Substituting, $ah_3 = bh_2h_1^{-1}h_3$. Since $h_2h_1^{-1}h_3 \in H$, we see $ah_3 \in bH$. Thus $aH \subseteq bH$. A similar argument shows $bH \subseteq aH$. Thus, $aH = bH$. $\square$

- Corollary In particular, $aH = eH = H \Leftrightarrow a \in H$ If $a \in H$, then $a \cdot e \in aH$, so $aH \cap eH \neq \emptyset$. So $aH = eH = H$.

- Lemma 2 Any two cosets have the same size. formally, if $a, b \in G$, then $|aH| = |bH|$.

  Proof Define $\Phi : aH \to bH$ by $\Phi(ah) = ba^{-1}ah = bh$. To see Φ is a bijection, we claim Φ^{-1} is given by $\Phi^{-1}(x) = ab^{-1}x$. We check $\Phi^{-1} \circ \Phi(x) = \Phi^{-1}(ba^{-1}x) = ab^{-1}ba^{-1}x = x$.
  Thus, $\Phi^{-1} \circ \Phi = id : aH \to aH$. Also, $\Phi \circ \Phi^{-1}(x) = \Phi(ab^{-1}x) = ba^{-1}ab^{-1}x = x$, so $\Phi \circ \Phi^{-1} = id : bH \to bH$.
  So, Φ^{-1} is a two-sided inverse, so Φ is a bijection between aH and bH, which means $|aH| = |bH|$. $\square$

- Proof of Lagrange's Theorem

  Let $\{a_1H, a_2H, \ldots, a_{[G:H]}H\}$ be the set of distinct left cosets of H in G.
  Every element of G is in exactly one of those cosets, i.e the set of left cosets of H partition G.
  Thus we have the disjoint union $G = a_1H \sqcup a_2H \sqcup \ldots \sqcup a_{[G:H]}H$. Thus $|G| = |a_1H| + |a_2H| + \ldots + |a_{[G:H]}H|$.

  from Lemma 1

  However, $|a_iH| = |eH| = |H|$ for every i, $1 \leq i \leq [G:H]$ by Lemma 2.
  Thus $|G| = \underbrace{|H| + |H| + \ldots + |H|}_{[G:H] \text{ times}}$. Therefore, $|G| = |H| \cdot [G:H]$. $\square$

- **Cor. 1** $[G:H] = \frac{|G|}{|H|}$. In particular, the order of a subgroup divides the order of a group.

- **Cor. 2** If $a \in G$, then $|a| \mid |G|$ $|a|$ divides $|G|$

  Proof $|a| = |\langle a \rangle|$ and Lagrange's Thm. says $|\langle a \rangle|$ divides $|G|$. $\square$

- **Cor. 3** A group of prime order is cyclic.

  - **Cor. 3+** If $|G| = p$ with p prime, then $G \cong \mathbb{Z}/p\mathbb{Z}$

  Proof

  Let $a \in G$ where $|G| = p$. Suppose $a \neq e$. Thus $|a| \neq 1$. But $|a|$ divides $|G|$. Thus $|a| = p$. Hence $G = \langle a \rangle$. $\square$

  Isomorphism $\varphi: a \longmapsto \bar{1}_p \in \mathbb{Z}/p\mathbb{Z}$, prove as exercise

- Converse of $|H| \mid |G|$ is not true. Just because $m \mid |G|$ doesn't mean we have a subgroup of order m Doesn't hold for general groups, but does hold for cyclic groups

  Ex A_4 has order 12, but has no subgroup of order 6.

- **Cor. 4** If $a \in G$, then $a^{|G|} = e$.

  Proof $|a| = |\langle a \rangle|$ divides $|G|$. Say $|G| = |a| \cdot k$ for some $k \in \mathbb{Z}^+$. Now $a^{|G|} = a^{|a| \cdot k} = (a^{|a|})^k = e^k = e$. $\square$

- **Cor. 5** (Fermat's Little Theorem)

  for every $a \in \mathbb{Z}$ and every prime p one has $a \equiv a^p \bmod p$.

  Proof. If $p \mid a$, then $0 \equiv 0 \bmod p$, and we are done.

  Now suppose $p \nmid a$. Thus $\bar{a} \in \mathbb{Z}/p\mathbb{Z}^\times$. Notice $|\mathbb{Z}/p\mathbb{Z}^\times| = |\{\bar{1}, \bar{2}, \ldots, \overline{p-1}\}| = p-1$.

  $a + p\mathbb{Z}$ coset

  By cor. 4, $\underline{(\bar{a})^{p-1} = \bar{1}}$. Hence, multiplying by $\bar{a}$, $(\bar{a})^p = \bar{a}$, thus $a^p \equiv a \bmod p$. $\square$

  also important result